

First, initiate N instances with a softmax distribution based on degree. Calculate instance fitness as the size of the clique. With a softmax-like tournament, perform the following:

Select $N/2$ pairs of mates to breed. Cross over. The crossover process is the following: If the intersection is not null, take the intersection. If the intersection is null, see if the intersection between their candidate sets is not null. If the intersection between the candidate sets is null, choose a partner at random

Then greedily add

If there

Generate an additional $N/2$ candidates by performing H-Opt.

OR

Select N pairs of mates to breed In-place softmax H-opt mutate O instances

Repeat.

Algorithm 1 Genetic Algorithm

- 1: **procedure** GENETICALGORITHM($G = (U, V, E), N, M$)
 - 2:
 - 3: **end procedure**
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